

Online Shopping Cart System - OOP Case Study

1. Introduction

This document demonstrates an Online Shopping Cart System implemented using Python with Object-Oriented Programming concepts.

Entities in the system:

Product – Represents store products

Cart – Stores items added by users

User – Represents a normal customer

Online Shopping Cart System – Project Structure

```
online_shopping_cart/  
├── main.py  
├── README.md  
├── models/  
│   ├── __init__.py  
│   ├── product.py  
│   ├── cart.py  
│   ├── user.py  
│   └── premium_user.py  
├── services/  
│   ├── __init__.py  
│   ├── cart_service.py  
│   └── store_service.py  
├── exceptions/  
│   ├── __init__.py  
│   └── cart_exceptions.py  
├── utils/  
│   ├── __init__.py  
│   └── file_handler.py  
└── data/  
    └── order_history.txt
```

```
└─ tests/
    └─ __init__.py
    └─ test_cart.py
```

Folder Explanation

main.py

Entry point of the application.

Responsibilities:

- Create products
- Create users
- Add/remove items from cart
- Display cart details

Example

```
from models.product import Product
from models.user import User

laptop = Product("Laptop",1000,5)
user = User("Alice")

user.cart.add_product(laptop,1)
```

models/

Contains **core business classes**.

```
models/
  product.py
  cart.py
  user.py
  premium_user.py
```

Classes inside

File	Class	Concept
product.py	Product	Encapsulation

cart.py	Cart	Cart operations
user.py	User	Base class

services/

Contains **business logic layer**.

```
services/
    cart_service.py
    store_service.py
```

Responsibilities:

- Manage product inventory
- Process cart operations
- Handle business rules

Example

```
store.add_product("Laptop",1000,5)
```

exceptions/

Contains **custom exceptions**.

```
exceptions/
    cart_exceptions.py
```

Custom exceptions:

```
class OutOfStockError(Exception):
    pass

class ProductNotFoundError(Exception):
    pass
```

utils/

Contains **utility functions**.

```
utils/
    file_handler.py
```

Handles **file operations**.

Example

```
def save_order(cart):  
    with open("data/order_history.txt","a") as f:  
        f.write(str(cart.view_cart()))
```

data/

Stores **application data files**.

```
data/  
    order_history.txt
```

Example content

```
Order1: Laptop x1, Mouse x2  
Order2: Keyboard x1
```

tests/

Contains **unit test cases**.

```
tests/  
    test_cart.py
```

Uses:

```
unittest
```

Example test

```
def test_add_product(self):  
    cart.add_product(laptop,1)  
    self.assertEqual(len(cart.items),1)
```

README.md

Contains:

- Project overview
- Setup instructions
- How to run
- How to run tests